Problems 1

1. $^{2}_{1}H$、$^{6}_{3}Li$、$^{10}_{5}B$、$^{14}_{7}N$

2. ①
$$\left.\begin{aligned}
&^{12}C + p \rightarrow {}^{13}N + \gamma \\
&^{13}N + p \rightarrow {}^{14}O + \gamma \\
&^{14}O \rightarrow {}^{14}N + e^{+} + \nu_e \\
&^{14}N + p \rightarrow {}^{15}O + \gamma \\
&^{15}O \rightarrow {}^{15}N + e^{+} + \nu_e \\
&^{15}N + p \rightarrow {}^{12}C + {}^{4}He
\end{aligned}\right\} \Rightarrow 4p \rightarrow {}^{4}He + 2e^{+} + 2\nu_e + 3\gamma$$

②
$$\begin{cases}
\text{理论应为 } {}^{13}N(p,\gamma){}^{14}O(,\beta^{+}\nu){}^{14}N \\
\text{实际应为 } {}^{13}N(,\beta^{+}\nu){}^{13}C(p,\gamma){}^{14}N
\end{cases}$$

区别为 ^{13}N 先衰变还是先俘获质子.

极高温下, 质子热运动能更大, 质子俘获反应的速率超越了原子核自发β衰变的速率.

3.

4. $N = \frac{m}{M} \times N_A \times \text{丰度} = \frac{1g}{39.10g/mol} \times 6.022\times10^{23}\ atoms/mol \times 0.0117\% = 1.802\times10^{18}$

$A_{total} = \frac{A_r}{10.67\%} = \frac{199}{0.1067}$ decays/min $= 1865$ decays/min

由 $A_{total} = \lambda N = \frac{\ln 2}{T_{\frac{1}{2}}} N$

$T_{\frac{1}{2}} = \frac{\ln 2 \cdot N}{A_{total}} = 6.697\times10^{14}$ min $\approx 1.273\times10\times10^{9}$ 年

文献值为 1.277×10^{9} y, 非常接近

5. $T_{Ni} = 5.9\ d$. $\lambda_{Ni} = \frac{\ln 2}{5.9} \approx 0.1175\ d^{-1}$

$T_{Co} = 77.3\ d$. $\lambda_{Co} = \frac{\ln 2}{77.3} \approx 0.008967\ d^{-1}$

$t=0$ 时. ^{56}Ni: N_0. ^{56}Co: 0.

$N_{Ni}(t) = N_0 \cdot e^{-\lambda_{Ni} \cdot t}$

$A_{Ni}(t) = \lambda_{Ni} N_{Ni}(t) = N_0 \lambda_{Ni} \cdot e^{-\lambda_{Ni} \cdot t}$

由贝特曼方程:

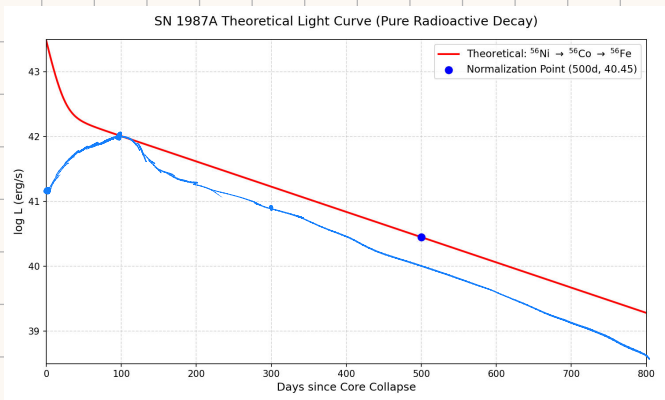
$N_{Co}(t) = N_0 \cdot \frac{\lambda_{Ni}}{\lambda_{Ni} - \lambda_{Co}} (e^{-\lambda_{Co} t} - e^{-\lambda_{Ni} \cdot t})$

$A_{Co}(t) = \lambda_{Co} \cdot N_{Co}(t)$

$A_{total}(t) = A_{Ni}(t) + A_{Co}(t)$, 定义 $A_{total}(t) = N_0 \cdot S(t)$

由 $L(t) = c \cdot A_{total}(t) \Rightarrow L(t) = c \cdot N_0 \cdot S(t)$, $K = c \cdot N_0$.

$\log_{10} L(t) = \log_{10} K + \log_{10} S(t)$. 由 $\log_{10} L(500) = 40.45 \Rightarrow K$.



相似点: 中后期均为指数衰减.

不同点: 早期: 理论曲线衰减, 实际曲线升

实际超新星外壳可能阻碍逃逸

晚期: 能量逸散使观测值小于理论值

代入数据编程绘图如右上:

公式为 $\log L(t) = 44.41 + \log_{10}[0.1078 \times \exp(-0.1175 \cdot t) + 0.00971 \times \exp(-0.00897 \cdot t)]$